

II coll 2023

Ex 1 $\beta_F = ?$ $\beta_R = 0.1$

$Q_1 = ?$

$V_O = -V_{CE2} = 4V$

$V_{CE2} = -4V$

$$R_P(I_{C3} + 2I_{C2}) = V_{CC} + V_{CE2}$$

$$R_P(I_{C3})(1 + 2) = V_{CC} + V_{CE2}$$

$$I_{C3} = I_{C2} = \frac{V_{CC} + V_{CE2}}{R_P(1 + 2)} = 1.06 \text{ mA}$$

$$I_{B4} = \frac{V_{CC} - V_O}{R_{E2}} - I_{C2} = \frac{V_{CC} - V_O}{R_{E2}} - I_{C2}\left(1 + \frac{1}{\beta_F}\right)$$

$$I_{C1} = I_{C3} + I_{C2} = I_{C3} + \frac{I_{C3}}{\beta_F}$$

$$I_{C3} + I_{C2} = I_{C1} = \beta_F I_{B4} = \beta_F \left[\frac{V_{CC} - V_O}{R_{E2}} - I_{C2}\left(1 + \frac{1}{\beta_F}\right) \right]$$

$$I_{C2} = \left[\frac{\beta_F V_{CC} - V_O}{R_{E2}} - I_{C1} \right] \cdot \frac{1}{\left(\frac{\beta_F + \beta_F}{\beta_F} + 1 \right)} = 0.86 \text{ mA}$$

$$I_{C1} = I_{C3} + I_{C2} = 1.066 \text{ mA}$$

$$V_{CE1} = V_O + V_{CE2} - R_{E1}(I_{C1})\left(1 + \frac{1}{\beta_F}\right) = 2.57 \text{ V} > V_{CEsat}$$

$$V_{CE2} = -4V < V_{CEsat}$$

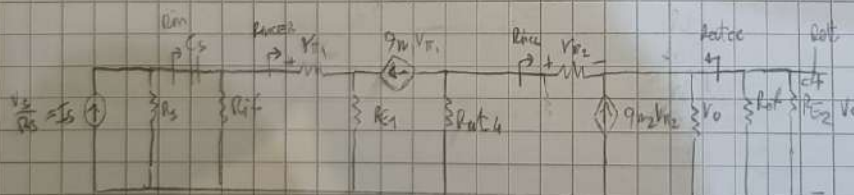
$$R_F = \left[\frac{V_O - \left(R_{E1} I_{C1} \left(1 + \frac{1}{\beta_F} \right) + V_{CE1} \right)}{I_{C1} / \beta_F} \right] = 482 \text{ k}\Omega$$

Small signal analysis

current mixing $\rightarrow$ short, no load equivalent
voltage sampling $\rightarrow$ short

V_O is transresistance
is amplifier topology

$$\begin{aligned} R_F &= R_F \\ R_F &= R_F \\ R_F &= R_F \\ R_F &= R_F \end{aligned}$$



$$g_{m1} = I_{C1} = 0.0412 \text{ A/V}$$

$$r_{\pi 1} = \frac{\beta_F}{g_{m1}} = \beta_F = 4.68 \text{ k}\Omega$$

$$g_{m2} = \frac{I_{C2}}{V_T} = 0.0323 \text{ A/V}$$

$$r_{\pi 2} = \frac{\beta_F}{g_{m2}} = \beta_F = 4.68 \text{ k}\Omega$$

$$A_F = \frac{A_{v2}}{1 + \beta_F A_{v1}} = -406.1 \text{ V/V}$$

$$A_V = \frac{V_O}{V_S} = A_F \cdot \frac{1}{R_S} = -40.61$$

$$R_{in}^{\alpha} = R_S || R_F || R_{in,CE1} = 9.163 \text{ k}\Omega$$

$$R_{in}^F = R_{in}^{\alpha} = 9.163 \text{ k}\Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}^F} + \frac{1}{R_S}} = 1.686 \text{ k}\Omega$$

$$R_{CE1} = V_O \quad A_V = \frac{V_O}{V_S} = ?$$

$$\alpha_{CC} = \frac{(1 + \beta_F)(R_{CE1} || R_{CE2} || V_O)}{R_{in,CC}} = 0.993 \text{ V/V}$$

$$R_{in,CC} = V_{T2} + (1 + \beta_F)(R_{CE1} || R_{CE2} || V_O) = 696.2 \text{ k}\Omega$$

$$\alpha_{CER} = -\beta_F(R_{CE1} || R_{CE2}) = -282.1 \text{ V/V}$$

$$R_{in,CE1} = r_{\pi 1} + (1 + \beta_F)R_{E1} = 141.56 \text{ k}\Omega$$

$$d_i = R_S || R_F || R_{in,CE1} = 9.163 \text{ k}\Omega$$

$$A^{\alpha} = \frac{V_O}{V_S} = d_i \alpha_{CER} \alpha_{CC} = -2.567 \text{ V/V}$$

$$R_{out} = R_{off} \parallel R_{E2} \parallel R_o \parallel R_{out2} = 1.34 \text{ K}\Omega$$

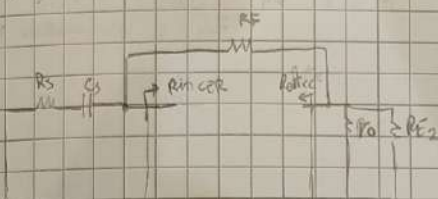
$$R_{out2} = \frac{-V_{R2} - \frac{V_{R2} R_{out1}}{R_{E2}}}{\frac{V_{R2} - g_{m2} V_{R2}}{R_{E2}}} = \frac{V_{R2} + R_{out1}}{1 - \beta_{R2}} = 1.875 \text{ M}\Omega$$

$$R_{outF} = \frac{R_{out2}}{1 + \beta_{R2}} = 212 \Omega$$

$$R_{out} = R_{outF}$$

$$\text{crossover } f_c = f_c = \frac{1}{2\pi C_S R_{S^{SC}}}$$

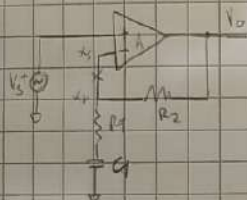
using SFC method feedback is reducing $R_{S^{SC}}$ so f_c is increasing $\rightarrow$ can't apply feedback method



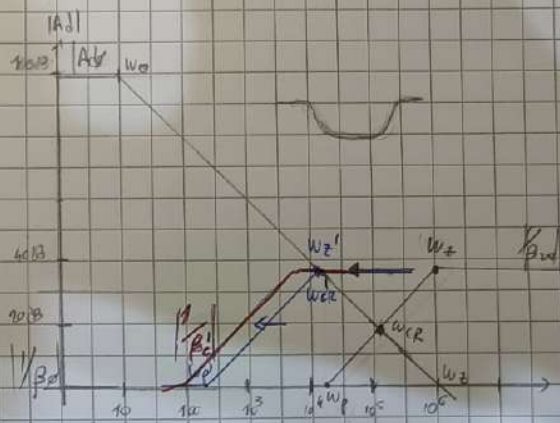
$$R_{S^{SC}} = R_S + R_{out2} \parallel (R_F + R_{out2} \parallel R_o \parallel R_{E2}) = 119.5 \text{ K}\Omega$$

$$f_c = \frac{1}{2\pi C_S R_{S^{SC}}} = 1.33 \text{ Hz}$$

Ex 2



$$A_d(s) = \frac{A_{d0}}{1 + s/\omega_0}$$



$$T = \frac{-V_o}{V_S} = A_d(s) \frac{R_1 + sC}{R_1 + \frac{1}{sC} + R_2}$$

$$R_z = \frac{1 + sRC}{1 + sC(R_1 + R_2)} = \frac{1 + s/\omega_z}{1 + s/\omega_p} \quad \text{pole and then zero}$$

$$\text{non-inverting configuration of opamp } A_v(s) = 1 + \frac{R_2}{R_1} = 1 + \frac{R_2}{R_1 + \frac{1}{sC}} = 1 + \frac{sCR_2}{1 + sCR_1} = \frac{1 + sC(R_1 + R_2)}{sCR_2} = A_v(s)$$

$$\omega_p = \frac{1}{C(R_1 + R_2)} = 14.49 \text{ Krad/s}$$

$$\beta_{d0} = 1 \rightarrow \frac{1}{\beta_{d0}} = 1 = 0 \text{ dB} \rightarrow (A\beta \approx \frac{1}{\beta} \text{ if } T \gg 1)$$

$$\omega_z = \frac{1}{CR_2} = 1 \text{ Mrad/s}$$

$$\beta_{d\infty} = \frac{R_1}{R_1 + R_2} \rightarrow \frac{1}{\beta_{d\infty}} = 1 + \frac{R_2}{R_1} = 69 = 37 \text{ dB}$$

$$\left\{ \begin{aligned} A_{d0} \omega_0 &= A_{\infty} \omega_{CR} \\ \frac{1}{\beta_{d0} \omega_p} &= \frac{A_{\infty}}{\omega_z} \end{aligned} \right\} \rightarrow A_{\infty} = \frac{\omega_{CR}}{\omega_p \beta_{d0}} \quad \omega_{CR} = \sqrt{\omega_p \beta_{d0} A_{d0} \omega_0} = 120.4 \text{ Krad/s}$$

$$PM = 180 - 20 \log \left(\frac{\omega_{CR}}{\omega_p} \right) - 20 \log \left(\frac{\omega_{CR}}{\omega_z} \right) + 20 \log \left(\frac{\omega_{CR}}{\omega_0} \right) = 13.7^\circ \quad \text{low phase margin}$$

$$\text{Since } PM = 60^\circ : \omega_{z'} = \frac{\omega_{CR}}{2} \quad \omega_{CR} \Rightarrow A_{d0} \omega_0 = \frac{1}{\beta_{d0}} \omega_{CR} \rightarrow \omega_{CR} = A_{d0} \beta_{d0} \omega_0 = 14.49 \text{ Krad/s} = \omega_p$$

$$\text{placed on the left before crossing} \quad \omega_{z'} = \frac{\omega_{CR}}{2} = 7.24 \text{ Krad/s} \rightarrow G = 1 = 138 \text{ dB}$$

$$PM = 63^\circ \quad \text{not } \omega_{z'} = \omega_{CR} \text{ because there's 4 poles later}$$